

LECTURE 13 SPURIOUS REGRESSION, TESTING FOR UNIT ROOT

Spurious regression

In this section, we consider the situation in which one unit root process, say Y_t , is regressed on another unit root process, say X_t , while the two processes are unrelated. Assume that

- $X_t = X_{t-1} + u_t$, $X_0 = 0$.
- $Y_t = Y_{t-1} + v_t$, $Y_0 = 0$.
- $\begin{pmatrix} u_t \\ v_t \end{pmatrix} = C(L) \varepsilon_t$.
- $\{\varepsilon_t\}$ is i.i.d., $E[\varepsilon_t] = 0$, $E[\varepsilon_t \varepsilon_t^\top] = \Sigma$ a positive definite matrix:

$$\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}.$$

- $\sum_{j=1}^{\infty} j^{1/2} \|C_j\| < \infty$, $C(1)$ is non-singular.

We can also set the initial values of X and Y to be different from zero: $X_0 = o_p(n^{1/2})$, and $Y_0 = o_p(n^{1/2})$.

Let

$$\begin{aligned} \Omega &= C(1) \Sigma C(1)^\top \\ &= \begin{pmatrix} \omega_u^2 & \omega_{uv} \\ \omega_{uv} & \omega_v^2 \end{pmatrix}, \end{aligned}$$

where ω_u^2 and ω_v^2 are the long-run variances of u_t and v_t respectively, and ω_{uv} is the long-run covariance between u_t and v_t .

Let $\hat{\beta}_n$ be the OLS regression coefficient:

$$\hat{\beta}_n = \frac{\sum_{t=1}^n X_t Y_t}{\sum_{t=1}^n X_t^2}.$$

Here we consider the regression without an intercept; however, essentially the same results can be obtained for the regression with an intercept.

Let

$$\begin{aligned} W_{X,n}(r) &= n^{-1/2} X_{[nr]}, \\ W_{Y,n}(r) &= n^{-1/2} Y_{[nr]}. \end{aligned}$$

From the FCLT we have

$$\begin{aligned} \begin{pmatrix} W_{X,n}(r) \\ W_{Y,n}(r) \end{pmatrix} &\Rightarrow \begin{pmatrix} B_X(r) \\ B_Y(r) \end{pmatrix} \\ &= \Omega^{1/2} \begin{pmatrix} W_X(r) \\ W_Y(r) \end{pmatrix}, \end{aligned}$$

where $W_X(r)$ and $W_Y(r)$ are two independent standard Brownian motions. Notice that convergence is joint, which is important for all subsequent results.

Consider first

$$\begin{aligned} n^{-3/2} \sum_{t=1}^n X_{t-1} &= n^{-1} \sum_{t=1}^n W_{X,n} \left(\frac{t-1}{n} \right) \\ &= n^{-1} \left(0 + W_{X,n} \left(\frac{1}{n} \right) + W_{X,n} \left(\frac{2}{n} \right) + \dots + W_{X,n} \left(\frac{n-1}{n} \right) \right). \end{aligned}$$

The function $W_{X,n}(r)$ is cadlag, constant on the interval $(t-1)/n \leq r < t/n$, and, therefore,

$$\begin{aligned} n^{-1} W_{X,n} \left(\frac{t-1}{n} \right) &= \int_{(t-1)/n}^{t/n} dr W_{X,n} \left(\frac{t-1}{n} \right) \\ &= \int_{(t-1)/n}^{t/n} W_{X,n}(r) dr, \end{aligned}$$

so that

$$\begin{aligned} 0 &= \int_0^{1/n} W_{X,n}(r) dr, \\ n^{-1} W_{X,n} \left(\frac{1}{n} \right) &= \int_{1/n}^{2/n} W_{X,n}(r) dr, \\ &\vdots \\ n^{-1} W_{X,n} \left(\frac{n-1}{n} \right) &= \int_{(n-1)/n}^1 W_{X,n}(r) dr. \end{aligned}$$

Hence,

$$\begin{aligned} n^{-3/2} \sum_{t=1}^n X_{t-1} &= \sum_{t=1}^n \int_{(t-1)/n}^{t/n} W_{X,n}(r) dr \\ &= \int_0^1 W_{X,n}(r) dr \\ &\rightarrow_d \int_0^1 B_X(r) dr. \end{aligned}$$

The last result follows from the CMT. Next,

$$\begin{aligned} n^{-3/2} \sum_{t=1}^n X_t &= n^{-3/2} \sum_{t=1}^n (X_{t-1} + u_t) \\ &= n^{-3/2} \sum_{t=1}^n X_{t-1} + n^{-3/2} \sum_{t=1}^n u_t \\ &= n^{-3/2} \sum_{t=1}^n X_{t-1} + o_p(1) \\ &\rightarrow_d \int_0^1 B_X(r) dr. \end{aligned}$$

Now, consider

$$\begin{aligned} n^{-2} \sum_{t=1}^n X_t^2 &= n^{-2} \sum_{t=1}^n (X_{t-1} + u_t)^2 \\ &= n^{-2} \sum_{t=1}^n X_{t-1}^2 + 2n^{-2} \sum_{t=1}^n X_{t-1} u_t + n^{-2} \sum_{t=1}^n u_t^2. \\ &= n^{-2} \sum_{t=1}^n X_{t-1}^2 + 2n^{-2} \sum_{t=1}^n X_{t-1} u_t + o_p(1). \end{aligned}$$

First,

$$\begin{aligned} n^{-2} \sum_{t=1}^n X_{t-1}^2 &= n^{-1} \sum_{t=1}^n \left(n^{-1/2} X_{t-1} \right)^2 \\ &= \int_0^1 W_{X,n}^2(r) dr \\ &\rightarrow_d \int_0^1 B_X^2(r) dr. \end{aligned}$$

Next,

$$2X_{t-1}u_t = X_t^2 - X_{t-1}^2 - u_t^2,$$

and

$$\begin{aligned} n^{-1} \sum_{t=1}^n X_{t-1}u_t &= n^{-1} \sum_{t=1}^n (X_t^2 - X_{t-1}^2 - u_t^2) / 2 \\ &= \left(n^{-1} X_n^2 - n^{-1} \sum_{t=1}^n u_t^2 \right) / 2 \\ &= \left(W_{X,n}^2(1) - n^{-1} \sum_{t=1}^n u_t^2 \right) / 2 \\ &\rightarrow_d (B_X^2(1) - \sigma_u^2) / 2 \\ &= (\omega_u^2 W_X^2(1) - \sigma_u^2) / 2, \end{aligned}$$

where $\sigma_u^2 = E[u_t^2]$. Therefore,

$$n^{-2} \sum_{t=1}^n X_{t-1}u_t = o_p(1),$$

and

$$n^{-2} \sum_{t=1}^n X_t^2 \rightarrow_d \int_0^1 B_X^2(r) dr. \quad (1)$$

Next, consider $\sum_{t=1}^n X_t Y_t$. First,

$$\begin{aligned} n^{-2} \sum_{t=1}^n X_{t-1}Y_{t-1} &= n^{-1} \sum_{t=1}^n \left(n^{-1/2} X_{t-1} \right) \left(n^{-1/2} Y_{t-1} \right) \\ &= \int_0^1 W_{X,n}(r) W_{Y,n}(r) dr \\ &\rightarrow_d \int_0^1 B_X(r) B_Y(r) dr. \end{aligned}$$

Next,

$$\begin{aligned} n^{-2} \sum_{t=1}^n X_t Y_t &= n^{-2} \sum_{t=1}^n (X_{t-1} + u_t) (Y_{t-1} + v_t) \\ &= n^{-2} \sum_{t=1}^n X_{t-1}Y_{t-1} + n^{-2} \sum_{t=1}^n X_{t-1}v_t + n^{-2} \sum_{t=1}^n Y_{t-1}u_t + n^{-2} \sum_{t=1}^n u_t v_t. \end{aligned}$$

Now, $n^{-1} \sum_{t=1}^n u_t v_t \rightarrow_p \sigma_{uv}$, where $\sigma_{uv} = E[u_t v_t]$, and, therefore,

$$n^{-2} \sum_{t=1}^n u_t v_t = o_p(1).$$

Further,

$$\begin{aligned}
n^{-2} \left| \sum_{t=1}^n Y_{t-1} u_t \right| &\leq n^{-2} \sqrt{\sum_{t=1}^n Y_{t-1}^2 \sum_{t=1}^n u_t^2} \\
&= n^{-1/2} \sqrt{n^{-2} \sum_{t=1}^n Y_{t-1}^2 n^{-1} \sum_{t=1}^n u_t^2} \\
&= o_p(1).
\end{aligned}$$

Similarly,

$$n^{-2} \sum_{t=1}^n X_{t-1} v_t = o_p(1).$$

Thus,

$$\begin{aligned}
n^{-2} \sum_{t=1}^n X_t Y_t &= n^{-2} \sum_{t=1}^n X_{t-1} Y_{t-1} + o_p(1) \\
&\rightarrow_d \int_0^1 B_X(r) B_Y(r) dr.
\end{aligned} \tag{2}$$

The convergence in distribution results in (1) and (2) are joint, and it follows that

$$\begin{aligned}
\hat{\beta}_n &\rightarrow_d \frac{\int_0^1 B_X(r) B_Y(r) dr}{\int_0^1 B_X^2(r) dr} \\
&= \xi.
\end{aligned}$$

The result holds even if $\{u_t\}$ and $\{v_t\}$ are independent. One might expect that $\hat{\beta}_n$ would converge in probability to zero; however, it converges in distribution to a random variable ξ and, therefore, is inconsistent. The random variable ξ can be interpreted as a regression coefficient from the "population" or "continuous time" regression of the Brownian motion B_Y against B_X .

Next, consider the usual t -statistic for $H_0 : \beta = 0$:

$$t_{\hat{\beta}_n} = \hat{\beta}_n / \left(\frac{s_n^2}{\sum_{t=1}^n X_t^2} \right)^{1/2},$$

where s_n^2 is the sample variance of the fitted residuals.

$$\begin{aligned}
s_n^2 &= (n-1)^{-1} \sum_{t=1}^n \left(Y_t - \hat{\beta}_n X_t \right)^2 \\
&= (n-1)^{-1} \left(\sum_{t=1}^n Y_t^2 + \hat{\beta}_n^2 \sum_{t=1}^n X_t^2 - 2\hat{\beta}_n \sum_{t=1}^n X_t Y_t \right) \\
&= \frac{n}{n-1} n^{-1} \left(\sum_{t=1}^n Y_t^2 - \hat{\beta}_n \sum_{t=1}^n X_t Y_t \right),
\end{aligned}$$

and, therefore,

$$\begin{aligned}
n^{-1} s_n^2 &\rightarrow_d \int_0^1 B_Y^2(r) dr - \xi \int_0^1 B_X(r) B_Y(r) dr \\
&= \int_0^1 (B_Y(r) - \xi B_X(r))^2 dr.
\end{aligned}$$

Lastly,

$$\begin{aligned} n^{-1/2} t_{\hat{\beta}_n} &= \hat{\beta}_n / \left(\frac{n^{-1} s_n^2}{n^{-2} \sum_{t=1}^n X_t^2} \right)^{1/2} \\ &\rightarrow_d \xi / \left(\frac{\int_0^1 (B_Y(r) - \xi B_X(r))^2 dr}{\int_0^1 B_X^2(r) dr} \right)^{1/2}. \end{aligned}$$

We conclude that

$$t_{\hat{\beta}_n} = O_p(n^{1/2}).$$

Hence, as $n \rightarrow \infty$, for any $K > 0$

$$\Pr(|t_{\hat{\beta}_n}| > K) \rightarrow 1,$$

and the econometrician will reject $H_0 : \beta = 0$ with probability approaching 1. This is despite the fact that the two variables X and Y can be independent.

Testing for unit root

Suppose that the scalar process $\{X_t\}$ satisfies the following assumptions:

- $X_t = \rho X_{t-1} + u_t$, $X_0 = 0$.
- $u_t = C(L) \varepsilon_t$.
- $\{\varepsilon_t\}$ is i.i.d., $E[\varepsilon_t] = 0$, $E[\varepsilon_t^2] = \sigma^2$.
- $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$, $C(1) \neq 0$.

We are interested in testing

$$H_0 : \rho = 1.$$

against

$$H_1 : |\rho| < 1.$$

Under the null, $X_t = I(1)$, while under the alternative, it is a stationary short memory process.

Consider the regression of X_t against X_{t-1} :

$$\begin{aligned} \hat{\rho}_n &= \frac{\sum_{t=1}^n X_{t-1} X_t}{\sum_{t=1}^n X_{t-1}^2} \\ &= \rho + \frac{\sum_{t=1}^n X_{t-1} u_t}{\sum_{t=1}^n X_{t-1}^2}. \end{aligned}$$

From the previous section, we know that

$$\begin{aligned} n^{-2} \sum_{t=1}^n X_{t-1}^2 &\rightarrow_d \int_0^1 B_X^2(r) dr \\ &= \omega_u^2 \int_0^1 W_X^2(r) dr, \end{aligned}$$

where W_X is a standard Brownian motion, and

$$\begin{aligned} n^{-1} \sum_{t=1}^n X_{t-1} u_t &\rightarrow_d \frac{1}{2} (B_X^2(1) - \sigma_u^2) \\ &= \frac{1}{2} (\omega_u^2 W_X^2(1) - \sigma_u^2) \\ &= \frac{\omega_u^2}{2} (W_X^2(1) - 1) + \lambda_u, \end{aligned}$$

where

$$\begin{aligned}
\lambda_u &= \frac{\omega_u^2 - \sigma_u^2}{2} \\
&= \frac{1}{2} \sum_{h=-\infty}^{\infty} (\gamma_u(h) - \sigma_u^2) \\
&= \sum_{h=1}^{\infty} \gamma_u(h).
\end{aligned}$$

Now, under $H_0 : \rho = 1$,

$$\begin{aligned}
n(\hat{\rho}_n - 1) &= \frac{n^{-1} \sum_{t=1}^n X_{t-1} u_t}{n^{-2} \sum_{t=1}^n X_{t-1}^2} \\
&\rightarrow_d \frac{\omega_u^2 (W_X^2(1) - 1) / 2 + \lambda_u}{\omega_u^2 \int_0^1 W_X^2(r) dr} \\
&= \frac{(W_X^2(1) - 1) / 2 + \lambda_u / \omega_u^2}{\int_0^1 W_X^2(r) dr}.
\end{aligned}$$

In the unit root case, the asymptotic distribution depends on functionals of a standard Brownian motion and the nuisance parameters, λ_u and ω_u^2 . The convergence rate of $\hat{\rho}_n$ is faster than the usual $\sqrt{n}$. Next, consider the t statistic for $H_0 : \rho = 1$.

$$T = (\hat{\rho}_n - 1) / \left(\frac{\hat{\sigma}_u^2}{\sum_{t=1}^n X_{t-1}^2} \right)^{1/2},$$

where

$$\begin{aligned}
\hat{\sigma}_u^2 &= n^{-1} \sum_{t=1}^n \hat{u}_t^2 \\
&= n^{-1} \sum_{t=1}^n (X_t - \hat{\rho}_n X_{t-1})^2 \\
&= n^{-1} \sum_{t=1}^n (X_t - X_{t-1} - (\hat{\rho}_n - 1) X_{t-1})^2 \\
&= n^{-1} \sum_{t=1}^n u_t^2 + n(\hat{\rho}_n - 1)^2 n^{-2} \sum_{t=1}^n X_{t-1}^2 - 2(\hat{\rho}_n - 1) n^{-1} \sum_{t=1}^n X_{t-1} u_t \\
&= n^{-1} \sum_{t=1}^n u_t^2 + O_p(n^{-1}) \\
&\rightarrow_p \sigma_u^2.
\end{aligned}$$

Thus,

$$\begin{aligned}
T &= n(\hat{\rho}_n - 1) / \left(\frac{\hat{\sigma}_u^2}{n^{-2} \sum_{t=1}^n X_{t-1}^2} \right)^{1/2} \\
&\rightarrow_d \left(\frac{(W_X^2(1) - 1) / 2 + \lambda_u / \omega_u^2}{\int_0^1 W_X^2(r) dr} \right) / \left(\frac{\sigma_u^2}{\omega_u^2 \int_0^1 W_X^2(r) dr} \right)^{1/2} \\
&= \frac{\omega_u}{\sigma_u} \frac{(W_X^2(1) - 1) / 2 + \lambda_u / \omega_u^2}{\left(\int_0^1 W_X^2(r) dr \right)^{1/2}}.
\end{aligned}$$

Again, the asymptotic distribution of the statistic depends on the unknown nuisance parameters λ_u , σ_u and ω_u . Phillips (1987) and Phillips and Perron (1988) suggested an adjustment, which leads to an asymptotic distribution free of nuisance parameters. Let $\hat{\sigma}_u^2$ and $\hat{\lambda}$ be consistent estimators of σ_u^2 and λ , where λ can be estimated using the Newey-West type estimator:

$$\hat{\lambda} = \sum_{h=1}^{m_n} \left(1 - \frac{h}{m_n + 1}\right) n^{-1} \sum_{t=h+1}^n \hat{u}_t \hat{u}_{t-h}.$$

Notice that a consistent estimator of the long-run variance ω_u^2 is

$$\hat{\omega}_u^2 = \hat{\sigma}_u^2 + 2\hat{\lambda}.$$

Consider the following modification of the t statistic.

$$Z_T = \frac{\hat{\sigma}_u}{\hat{\omega}_u} T - \frac{\hat{\lambda}}{\hat{\omega}_u (n^{-2} \sum_{t=1}^n X_{t-1}^2)^{1/2}}.$$

Under $H_0 : \rho = 1$,

$$Z_T \rightarrow_d \frac{1}{2} \frac{W_X^2(1) - 1}{\left(\int_0^1 W_X^2(r) dr\right)^{1/2}}.$$

Under the alternative, $|\rho| < 1$, and $\hat{\rho}_n - 1$ converges in probability to a negative constant. Consequently, under the stationary alternatives, T and Z_T diverge to $-\infty$. One should reject the null of unit root when

$$Z_T < c_\alpha,$$

where c_α is such that

$$\Pr(Z_T < c_\alpha) \xrightarrow{H_0: \rho=1} \alpha$$

Under the null, the distribution is non-standard. However, it is parameter-free, and the critical values can be simulated as follows. First, one generates n independent $N(0, 1)$ random variables $u_{1,r}^*, \dots, u_{n,r}^*$ and computes

$$Z_{T,r}^* = \frac{1}{2} \frac{(n^{-1/2} \sum_{t=1}^n u_{t,r}^*)^2 - 1}{\left(n^{-2} \sum_{t=1}^n \left(\sum_{s=1}^t u_{s,r}^*\right)^2\right)^{1/2}}.$$

One repeats this for $r = 1, \dots, R$, where R is large. The simulated critical value $c_{\alpha,R}$ is the α quantile of $\{Z_{T,1}^*, \dots, Z_{T,R}^*\}$.

While the distribution of Z_T is free of nuisance parameters, it depends on the model. For example, in general, one would like to allow for an intercept, $X_t = \mu + X_{t-1} + u_t$. In this case, $\hat{\rho}_n$ depends on the demeaned X_t :

$$\begin{aligned} \hat{\rho}_n &= \frac{\sum_{t=1}^n (X_{t-1} - \bar{X}_n) X_t}{\sum_{t=1}^n (X_{t-1} - \bar{X}_n)^2}, \text{ where} \\ \bar{X}_n &= n^{-1} \sum_{t=1}^n X_{t-1}. \end{aligned}$$

Notice that

$$\begin{aligned}
n^{-2} \sum_{t=1}^n (X_{t-1} - \bar{X}_n)^2 &= \int_0^1 \left(W_{X,n}(r) - \int_0^1 W_{X,n}(r) dr \right)^2 dr \\
&\rightarrow_d \int_0^1 \left(B_X(r) - \int_0^1 B_X(r) dr \right)^2 dr \\
&= \omega_u^2 \int_0^1 \left(W_X(r) - \int_0^1 W_X(s) ds \right)^2 dr \\
&= \omega_u^2 \int_0^1 \widetilde{W}_X^2(r) dr,
\end{aligned}$$

where $\widetilde{W}_X$ is a demeaned standard Brownian motion:

$$\widetilde{W}_X(r) = W_X(r) - \int_0^1 W_X(s) ds.$$

Hence, in this case,

$$\begin{aligned}
T &= (\hat{\rho}_n - 1) / \left(\frac{\hat{\sigma}_u^2}{\sum_{t=1}^n (X_{t-1} - \bar{X}_n)^2} \right)^{1/2}, \\
Z_T &= \frac{\hat{\sigma}_u}{\hat{\omega}_u} T - \frac{\hat{\lambda}}{\hat{\omega}_u \left(\sum_{t=1}^n (X_{t-1} - \bar{X}_n)^2 \right)^{1/2}},
\end{aligned}$$

and the asymptotic distribution of Z_T under the null of unit root is given by

$$\frac{1}{2} \frac{\widetilde{W}_X^2(1) - 1}{\left(\int_0^1 \widetilde{W}_X^2(r) dr \right)^{1/2}}.$$